

RESTING MEMBRANE POTENTIAL & ACTION POTENTIAL

NotesGPT-Style High-Yield Summary

SOURCE SCOPE Built from the provided 39-slide membrane-potential/action-potential deck. No textbook file was attached in this request, so no textbook-only material has been added.

Organization: follows the slide sequence from ion distribution -> resting membrane potential -> local/action potentials -> ion-channel gating and propagation -> cardiac and smooth-muscle action potentials.

1. Ionic Distribution & Resting Membrane Potential

Slides 2-14: the electrochemical foundation of membrane voltage

Unequal ion distribution across ECF and ICF

- **Extracellular fluid (ECF)** contains a large amount of **Na⁺** and only a small amount of **K⁺**; **intracellular fluid (ICF)** shows the opposite pattern.

WHY: A membrane voltage can only arise if ions are not distributed equally on the two sides of the membrane.

- **Cl⁻** is concentrated mainly in ECF, while **phosphates and proteins** are much more concentrated in ICF.

WHY: These unequal distributions create different chemical environments on the two sides of the membrane.

- The slide comparison gives representative concentrations such as **Na⁺ 142 mEq/L outside vs 10 mEq/L inside, K⁺ 4 vs 140 mEq/L, and Cl⁻ 103 vs 4 mEq/L.**

WHY: The large Na⁺ and K⁺ gradients make these ions especially important when the membrane becomes permeable to them.

- The figure also emphasizes very low intracellular **Ca⁺⁺** relative to extracellular Ca⁺⁺ and higher intracellular **phosphate/protein** content.

WHY: The figure is meant to show that each ion has its own concentration gradient rather than all solutes following one common pattern.

Species	ECF (slide figure)	ICF (slide figure)
Na ⁺	142 mEq/L	10 mEq/L
K ⁺	4 mEq/L	140 mEq/L
Ca ⁺⁺	2.4 mEq/L	0.0001 mEq/L
Mg ⁺⁺	1.2 mEq/L	58 mEq/L
Cl ⁻	103 mEq/L	4 mEq/L
HCO ₃ ⁻	28 mEq/L	10 mEq/L
Phosphates	4 mEq/L	75 mEq/L
Proteins	2 g/dL (5 mEq/L)	16 g/dL (40 mEq/L)

Definition and range of resting membrane potential (RMP)

- **Resting membrane potential** is the potential difference across the membrane of an **inactive cell**, with the **inside negative relative to the outside**.

WHY: RMP describes the resting electrical separation of charge across the membrane, not active signaling.

- RMP ranges roughly from **-10 to -100 mV** depending on the cell. The slides list a typical **neuron at -70 mV, skeletal muscle at -90 mV, and a typical epithelial cell at less than -50 mV.**

WHY: Different cell types have different membrane permeabilities and ion distributions, so they settle at different resting voltages.

Two requirements for a membrane potential

- A potential difference requires **(1) an unequal distribution of one or more ions across the membrane and (2) membrane permeability to one or more of those ions.**

WHY: A concentration gradient provides the driving force, while permeability gives ions a path to move and separate charge.

Nernst potential and Nernst equation

- The **Nernst potential (equilibrium potential)** for an ion is the diffusion potential that **exactly opposes net diffusion** of that ion through the membrane.

WHY: At this voltage, the electrical force opposing ion movement balances the concentration-driven tendency for that ion to diffuse.

- The general slide equation is $E = (RT / zF) \log([ion]_o / [ion]_i)$. For the simplified 37 C form shown, monovalent ions use **$\pm 61 \log([in]/[out])$ mV**; for Ca^{++} , the factor is **61/2**.

WHY: The charge on the ion changes how strongly voltage affects it, which is why divalent Ca^{++} uses half the monovalent factor.

- The slide instructs: use **- for cations** and **+ for anions** in the simplified $[in]/[out]$ form.

WHY: The sign convention accounts for whether positive or negative charge is being moved across the membrane.

- Memorized equilibrium potentials given in the deck: **Na^+ about +70 mV**, **K^+ about -94 mV**, and **Ca^{++} about +132 mV**.

WHY: These values show the membrane voltage toward which each ion would drive the cell if that ion dominated membrane permeability.

Ionic basis and origin of normal RMP

- **Na^+ is actively transported out** of cells and **K^+ into** cells, but at rest there are **more K^+ channels**, so **K^+ permeability is greater than Na^+ permeability**.

WHY: Because K^+ has the larger resting permeability, K^+ movement dominates the resting voltage.

- The deck separates the normal RMP into contributions from **K^+ diffusion**, **Na^+ diffusion**, and the **Na^+-K^+ pump**.

WHY: RMP is not a single-process phenomenon; diffusion through channels and active transport contribute in different ways.

- K^+ and Na^+ diffusion alone produce about **-86 mV**, with almost all of that determined by K^+ diffusion; the continuously acting **Na^+-K^+ pump adds about -4 mV**, giving about **-90 mV**.

WHY: This shows why K^+ diffusion is the major immediate determinant, while the pump makes a smaller direct electrical contribution.

- The slide experiment with **ouabain**, an inhibitor of the Na^+-K^+ pump, showed that membrane potential can persist for **a few hours even when the pump stops**.

WHY: If the pump directly created the entire membrane voltage, blocking it would abolish the voltage immediately; persistence shows that stored ion gradients and membrane permeability are central.

- The **Na^+-K^+ pump is nevertheless essential for maintaining the K^+ and Na^+ concentration gradients** that support RMP over time.

WHY: The channels use existing gradients to generate voltage, while the pump preserves those gradients so they do not dissipate.

Goldman-Hodgkin-Katz (GHK) concept

- When both **Na^+ and K^+ are permeant**, membrane potential can be approximated by the **GHK equation**, which weights the concentrations of each ion by its **membrane permeability**.

WHY: Unlike the Nernst equation for one ion, GHK combines the influence of more than one permeant ion.

- The slide example states **P_K is 25 times P_{Na}** , so **$P_{Na}/P_K = 0.04$** ; substituting the shown Na^+ and K^+ concentrations gives **$V_m = -70.6$ mV**.

WHY: Because K^+ is much more permeable than Na^+ , the combined membrane voltage lies much closer to the K^+ equilibrium tendency than to the Na^+ tendency.

CORE RMP LOGIC High intracellular K^+ + a membrane that is much more permeable to K^+ at rest $\rightarrow K^+$ leaves through K^+ channels while intracellular negatively charged proteins cannot follow $\rightarrow$ the inside becomes

negative. The Na⁺-K⁺ pump maintains the gradients and contributes a smaller direct electrical effect.

Concept	High-yield point	Slide-supported number/example
RMP	Inside negative relative to outside	Neuron -70 mV; skeletal muscle -90 mV
Nernst potential	Voltage that exactly opposes net diffusion of one ion	Na ⁺ +70; K ⁺ -94; Ca ⁺⁺ +132 mV
Resting permeability	K ⁺ permeability > Na ⁺ permeability	PK ≈ 25 x PNa in GHK example
Diffusion contribution	Mostly K ⁺ -determined	K ⁺ + Na ⁺ diffusion ≈ -86 mV
Na ⁺ -K ⁺ pump	Maintains gradients; small direct voltage effect	≈ -4 mV direct contribution

2. Excitability, Local Potentials & Nerve Action Potentials

Slides 15-24: how excitable cells transition from resting voltage to an action potential

Excitable cells and types of excitation

- **Excitable cells** listed in the slides include **neurons, muscle cells, some endocrine cells, and some other cells such as macrophages**. Electrical, chemical, or mechanical stimulation can change their membrane potential dramatically for short periods (milliseconds).

WHY: Excitability means the membrane voltage can respond rapidly to a stimulus instead of remaining fixed at its resting value.

- Two broad forms of excitation are shown: **local, non-propagated potentials** (synaptic, generator, or electrotonic potentials) and **propagated potentials (action potentials)**.

WHY: The key distinction is whether the voltage change stays local or regenerates along the membrane.

Local potentials

- Local potentials can produce **depolarization or hyperpolarization**, and their size can vary with stimulus strength.

WHY: Local responses are graded rather than fixed in size.

- Local potentials can be **summed by temporal summation and spatial summation**.

WHY: Multiple local voltage changes can add together either across time or from different locations.

- The slide graph shows local potentials decreasing with distance from the stimulus site, in contrast with the non-decremental propagation later emphasized for action potentials.

WHY: A local potential is not regenerated at each segment of membrane, so its size can fade as it spreads.

Basic nerve action potential sequence

- A nerve **action potential (AP)** begins with a sudden shift from the normal negative resting potential toward a positive potential (**depolarization**), followed by return toward the negative potential (**repolarization**).

WHY: The AP is a rapid, transient reversal and restoration of membrane voltage.

- During **depolarization**, the membrane suddenly becomes highly permeable to Na^+ , and Na^+ influx neutralizes the negative resting state.

WHY: Na^+ entering the cell carries positive charge inward, driving membrane voltage upward.

- In large nerve fibers, Na^+ influx can produce an **overshoot beyond 0 mV**; the slide notes that in small neurons and many CNS neurons the voltage may reach only around zero.

WHY: The extent of inward positive charge determines whether the membrane becomes truly positive or merely loses its negative voltage.

- During **repolarization**, Na^+ channels begin to close and K^+ channels open more than normal; rapid K^+ diffusion outward re-establishes the negative membrane potential.

WHY: Stopping Na^+ entry removes the depolarizing drive while K^+ leaving carries positive charge out, returning the inside toward negativity.

After-potentials / hyperpolarization

- The sharp rise and rapid fall form the neuronal **spike potential**.

WHY: This is the fast central event of the nerve action potential tracing.

- The slower fall at the end is labeled **after-depolarization (negative after-potential)** in the slide.

WHY: The tracing does not always return instantly to the exact resting level after the main spike.

- A small overshoot in the hyperpolarizing direction with a prolonged tracing is labeled **after-hyperpolarization (positive after-potential)**.

WHY: The membrane can transiently become more negative than its resting level before fully settling back.

Initiation and threshold

- AP initiation depends on a **positive-feedback cycle involving Na⁺ channels**.

WHY: Once Na⁺ entry depolarizes the membrane enough, that depolarization promotes further Na⁺ channel opening and even more Na⁺ entry.

- The slides define **threshold** as the point at which Na⁺ entering becomes greater than K⁺ leaving. A sudden rise of about **15-30 mV** is usually required, with threshold values shown in the approximate **-50 to -70 mV** range.

WHY: Threshold marks the transition from a small local response to a self-amplifying depolarization.

- The threshold-for-excitation graph emphasizes both **stimulus intensity and duration**: subthreshold stimuli produce only local responses, whereas a stimulus that reaches threshold triggers a full AP.

WHY: A stimulus must alter membrane voltage enough, for long enough, to start the regenerative Na⁺ feedback loop.

Feedback diagrams from the slides

- **Na⁺ positive feedback**: depolarizing stimulus → opening of voltage-gated Na⁺ channels → increased PNa → increased Na⁺ influx → further membrane depolarization.

WHY: Each step reinforces the next, so the depolarization accelerates once threshold is crossed.

- The Na⁺ positive-feedback loop is stopped by **inactivation of Na⁺ channels**.

WHY: Without termination, the regenerative cycle would not end and repolarization could not proceed.

- **K⁺ negative feedback**: Na⁺-driven depolarization → opening of voltage-gated K⁺ channels → increased PK → increased K⁺ efflux → repolarization.

WHY: K⁺ efflux reverses the voltage change that opened the K⁺ channels, so this loop counteracts depolarization rather than reinforcing it.

Feature	Local potential	Action potential
Propagation	Local / non-propagated	Propagated
Amplitude	Varies with stimulus	Triggered once threshold is exceeded
Summation	Temporal and spatial summation	Not described as summing in these slides
Distance	Can decline with distance	Propagates without decrement
Examples	Synaptic, generator, electrotonic	Nerve AP

3. Voltage-Gated Channels, Refractory Periods & Propagation

Slides 25-33: channel gating explains the AP shape, refractoriness, and conduction

Voltage-gated Na⁺ channel: activation and inactivation

- When membrane potential reaches roughly **-70 to -50 mV**, the Na⁺ channel **activation gate opens rapidly**, allowing Na⁺ to pour inward.

WHY: The activation gate is the fast step that launches the rising phase once threshold is reached.

- The same voltage change also drives the **inactivation gate toward the closed state**, but this conformational change is **slower** than activation.

WHY: The delay allows Na⁺ influx to occur briefly before the channel becomes inactivated.

- The slide diagram shows Na⁺ channels cycling through **resting -> activated -> inactivated** states as membrane voltage rises and then returns.

WHY: The sequence explains why a voltage-gated Na⁺ channel can open quickly but then become temporarily unavailable even while the membrane is still depolarized.

Voltage-gated K⁺ channel

- As membrane potential rises from about **-90 mV toward zero**, voltage-gated K⁺ channels undergo a **slow conformational opening** that increases K⁺ diffusion outward.

WHY: The slower timing means K⁺ current becomes prominent after Na⁺-driven depolarization has already begun.

- K⁺ channel opening occurs at about the same time as **Na⁺ channel inactivation**.

WHY: The depolarizing Na⁺ current is therefore being shut down while the repolarizing K⁺ current is increasing.

Re-establishing Na⁺ and K⁺ gradients

- The **Na⁺-K⁺ pump** re-establishes the normal Na⁺ and K⁺ ionic gradients after activity; the slide states its activity may increase **up to 8-fold after an AP**.

WHY: The AP moves only a fraction of the available ions, but repeated activity still requires active transport to preserve the long-term gradients.

- The pump diagram shows **3 Na⁺ moved out and 2 K⁺ moved in** using ATP.

WHY: This is the transport pattern used by the membrane to restore and maintain the normal Na⁺/K⁺ distribution.

Absolute and relative refractory periods

- During the **absolute refractory period**, **no stimulus, regardless of strength, can excite the nerve**.

WHY: The membrane is temporarily unable to generate another AP during this interval.

- During the **relative refractory period**, a **stronger-than-normal stimulus** can trigger excitation.

WHY: Excitability has begun to recover but has not yet returned to the normal resting level.

- The integrated slide graph links refractoriness to Na⁺/K⁺ channel states: excitability falls to **zero** during the absolute refractory period and then **gradually increases** during the relative refractory period.

WHY: Recovery of channel availability occurs progressively rather than instantaneously.

- In the same graph, Na⁺ **permeability rises sharply and falls early**, whereas K⁺ **permeability rises later and remains elevated longer**, matching depolarization followed by repolarization/hyperpolarization.

WHY: The timing of each ion permeability change mirrors the direction of the membrane-voltage change it produces.

Core properties of the action potential

- Depolarization must **exceed threshold** to trigger an AP.

WHY: Subthreshold voltage changes remain local; crossing threshold starts the regenerative event.

- The AP is **all-or-none**.

WHY: Once threshold is reached, the AP is generated as a full event rather than as a partially sized spike.

- The AP **propagates without decrement**.

WHY: Each new membrane segment regenerates the AP rather than merely receiving a fading passive voltage change.

Continuous and saltatory conduction

- In **continuous conduction**, the AP travels in **all directions away from the stimulus**, including along branches of a nerve fiber, until the membrane has depolarized.

WHY: Adjacent membrane regions are successively brought to threshold so the excitation spreads along the fiber.

- In **saltatory conduction**, depolarization occurs at the **nodes of Ranvier** rather than along every part of the myelinated axon.

WHY: The slide emphasizes that restricting active depolarization to nodes changes how the impulse advances along a myelinated fiber.

- Saltatory conduction **increases nerve-transmission velocity** and **conserves axonal energy because only the nodes depolarize**.

WHY: Fewer membrane regions undergo the full ionic current sequence, so transmission is both faster and less energetically demanding according to the slide.

AP phase / interval	Dominant slide event	Functional consequence
Threshold / start	Rapid voltage-gated Na ⁺ activation	Regenerative depolarization begins
Depolarization	Na ⁺ influx; P _{Na} rises sharply	Membrane potential moves positive
Early repolarization	Na ⁺ inactivation + delayed K ⁺ opening	Depolarizing current falls while outward K ⁺ current rises
Repolarization / after-potential	K ⁺ efflux remains elevated	Membrane returns negative; may transiently hyperpolarize
Absolute refractory	Excitability = zero in integrated graph	No stimulus can trigger another AP
Relative refractory	Excitability gradually recovers	Stronger-than-normal stimulus can trigger an AP

INTEGRATED CHANNEL LOGIC Fast Na⁺ activation gives the steep upstroke. Slower Na⁺ inactivation plus delayed K⁺ activation terminate the upstroke and produce repolarization. The timing of channel recovery creates the absolute and relative refractory periods.

4. Cardiac Action Potentials

Slides 34-38: ventricular/Purkinje, nodal, and atrial electrical patterns

Ventricular myocardium / Purkinje fibers

- The ventricular/Purkinje action potential is divided into **phases 0-4**.

WHY: The phase system organizes the sequential ionic events across one cardiac action potential.

- Phase 0: rapid Na⁺ influx.**

WHY: Rapid inward positive current produces the steep depolarizing upstroke shown in the tracing.

- Phase 1: described as a small/insignificant phase with K⁺ out.**

WHY: The slide treats this as a brief transition before the plateau.

- Phase 2 (plateau): Ca⁺⁺ in while K⁺ moves out.**

WHY: Opposing inward Ca⁺⁺ and outward K⁺ currents are shown during the prolonged plateau portion of the tracing.

- Phase 3: rapid repolarization with K⁺ out.**

WHY: Outward K⁺ movement is the ionic event emphasized for returning the membrane toward its resting level.

- Phase 4: ions are returned to the proper side between beats; the slide lists the ATPase pump, Na⁺-Ca⁺⁺ exchanger, and Ca⁺⁺ pump.**

WHY: The interval between action potentials is used to restore the ion distributions required for the next cycle.

Nodal tissue: the slow, self-activating action potential

- Nodal tissue has a **"slow" action potential** and is **self-activating through diastolic depolarization**.

WHY: The membrane does not remain at one flat resting voltage; it drifts toward threshold on its own.

- The slides attribute diastolic depolarization to being **leaky to Na⁺ at rest**, producing **autorhythmicity**.

WHY: A gradual inward positive current moves the membrane toward the threshold for the next action potential.

- The slide shows nodal membrane potential already around **-70 mV** (with a note that Guyton gives **-55 mV**) and states gates open fully around **-40 mV**.

WHY: Because the starting voltage is closer to threshold than the -90 mV ventricular level, less depolarization is needed to reach the firing threshold shown.

- Phase 0 is mostly Ca⁺⁺ influx**, so the upstroke is **slow**.

WHY: The slide directly links the dominant inward ion in phase 0 with the slower nodal depolarization.

- Phase 3 is K⁺ efflux**, as in skeletal and nerve tissue.

WHY: Outward K⁺ movement again provides the repolarizing current.

Atrial action potential

- The **atrial action potential is very similar to the ventricular myocardial action potential**.

WHY: Atria and ventricles use a comparable phase pattern in the figure.

- The atrial AP is **smaller and shorter in duration**, with a **shorter refractory period**, especially a shorter **phase 2**.

WHY: The figure highlights a reduced plateau and earlier completion of the atrial waveform compared with the ventricular tracing.

Electrical vs mechanical timing: skeletal vs cardiac muscle

- In the slide comparison, the **skeletal-muscle action potential is very brief** compared with the longer mechanical tension response.

WHY: The electrical event ends early while the plotted contraction continues for much longer.

- In cardiac muscle, the **action potential is prolonged**, and the **absolute refractory period (ARP)** occupies most of the time during which the mechanical response develops; a shorter **relative refractory period (RRP)** follows.

***WHY:** The cardiac electrical and mechanical events overlap much more extensively than the skeletal tracing in the provided figure.*

Tissue / phase pattern	Key ionic or timing feature in slides	Distinctive point
Ventricle / Purkinje	0 Na ⁺ in; 1 K ⁺ out; 2 Ca ⁺⁺ in + K ⁺ out; 3 K ⁺ out; 4 ion restoration	Long plateau phase
Nodal tissue	Diastolic depolarization; phase 0 mostly Ca ⁺⁺ ; phase 3 K ⁺ out	Slow, self-activating / autorhythmic
Atrium	Similar to ventricle	Smaller, shorter AP; shorter phase 2 and refractory period
Skeletal muscle timing	Brief AP compared with contraction	Electrical spike ends well before mechanical response ends
Cardiac muscle timing	Prolonged AP + long ARP	Electrical/refractory period overlaps most mechanical response

5. Smooth-Muscle Action-Potential Patterns

Slide 39: three electrical patterns shown in smooth muscle

- **A. Typical smooth-muscle spike potential:** a single action potential elicited by an **external stimulus**.
WHY: This tracing shows a discrete stimulus-triggered electrical event.
- **B. Repetitive spike potentials:** elicited by **slow rhythmical electrical waves** that occur spontaneously in smooth muscle of the **intestinal wall**.
WHY: The slow-wave baseline periodically reaches levels that trigger bursts of spikes in the figure.
- **C. Action potential with a plateau:** recorded from a smooth-muscle fiber of the **uterus**.
WHY: The tracing remains depolarized for a sustained interval rather than returning immediately after a narrow spike.

Pattern	Trigger / setting shown	Tracing appearance
Typical spike	External stimulus	Single brief spike
Repetitive spikes	Spontaneous slow rhythmical waves in intestinal wall	Bursts of spikes riding on slow waves
Plateau AP	Uterine smooth-muscle fiber	Prolonged depolarized plateau before repolarization

Rapid Final Review

A slide-order consolidation for memorization

- **RMP needs two things:** an ion concentration gradient + permeability to that ion.
WHY: A gradient without a pathway cannot move charge; a pathway without a gradient has no net chemical driving force.
- **At rest, K⁺ permeability dominates Na⁺ permeability**, so K⁺ diffusion is the major determinant of RMP.
WHY: The membrane voltage is pulled most strongly toward the equilibrium tendency of the ion with the largest permeability.
- **The Na⁺-K⁺ pump maintains Na⁺/K⁺ gradients and adds a smaller direct negative contribution to RMP.**
WHY: The ouabain experiment shows the existing membrane potential persists temporarily even when the pump stops.
- **Na⁺ activation is fast, Na⁺ inactivation is delayed, and K⁺ activation is slow/delayed.**
WHY: That timing creates the upstroke, terminates Na⁺ entry, and drives repolarization.
- **Absolute refractory:** impossible to excite. **Relative refractory:** stronger-than-normal stimulus can excite.
WHY: Excitability first falls to zero and then progressively recovers.
- **Action potentials are threshold-dependent, all-or-none, and propagate without decrement.**
WHY: A regenerated AP behaves differently from a graded local potential.
- **Saltatory conduction** increases velocity and conserves energy because only **nodes** depolarize.
WHY: The myelinated axon does not need the full ionic depolarization sequence at every membrane segment.
- **Ventricular/Purkinje cardiac AP: 0 Na⁺ in -> 1 K⁺ out -> 2 Ca⁺⁺ in + K⁺ out -> 3 K⁺ out -> 4 ion restoration.**
WHY: The phase sequence summarizes the ionic basis of the ventricular cardiac tracing.

- Nodal AP: **diastolic depolarization**, phase 0 mainly **Ca⁺⁺ in**, phase 3 **K⁺ out**.

WHY: This pattern explains the slow, self-activating nodal waveform shown in the slides.

- Smooth muscle may show a **single spike, repetitive spikes on slow waves, or a plateau AP**.

WHY: The final slide emphasizes that smooth-muscle electrical behavior is not represented by one universal waveform.

[End of Notes]